Final Exam - Applied Abstract Algebra

June 9th, 2026

First Problem Variants

Variant 1. *(40 points)* Let Q be a quiver:

$$\begin{tikzcd}
 & & & 4 \\
1 \arrow[r] & 2 & 3 \arrow[l] \arrow[ur] \arrow[dr] & \\
 & & & 5
\end{tikzcd}$$

a). Construct $P(3)$ and $I(2)$ and using projective and injective resolution calculate $\tau(P(3))$ and $\tau^{-1}(I(2))$, where τ denotes the Auslander-Reiten translation.

b). Construct a triangulation of punctured polygon corresponding to Q and find which arcs with respect to this triangulation correspond to $P(3)$, $I(2)$, $\tau(P(3))$, $\tau^{-1}(I(2))$ and to indecomposable representation of dimension vector $(1, 2, 2, 1, 1)$.

Proof. The projective $P(i)$ is spanned by paths starting at i, and the injective $I(i)$ is spanned by paths ending at i. Hence

$$\dim P(3) = (0, 1, 1, 1, 1), \qquad \dim I(2) = (1, 1, 1, 0, 0).$$

The nonzero maps in $P(3)$ are the identity maps on the arrows $3 \to 2$, $3 \to 4$, and $3 \to 5$; the nonzero maps in $I(2)$ are the identity maps on $1 \to 2$ and $3 \to 2$.

Since $P(3)$ is projective, its minimal projective resolution is $0 \to 0 \to P(3) \to P(3) \to 0$, so applying the Nakayama definition of Auslander-Reiten translation gives $\tau P(3) = 0$. Similarly, $I(2)$ is injective, so $\tau^{-1} I(2) = 0$.

Let $T_Q = \{\gamma_1, \ldots, \gamma_5\}$ be the triangulation of the punctured pentagon obtained from Q by Schiffler's construction, with γ_i corresponding to the vertex i. Let R denote elementary clockwise rotation, with tag change at the puncture. Then

$$P(3) \leftrightarrow R^{-1}\gamma_3, \qquad I(2) \leftrightarrow R\gamma_2.$$

The actual objects $\tau P(3)$ and $\tau^{-1} I(2)$ are zero in $\operatorname{rep} Q$; formally rotating the arcs gives the boundary triangulation arcs γ_3 and γ_2, respectively.

The indecomposable representation of dimension vector $(1, 2, 2, 1, 1)$ may be written as

$$X_1 = X_4 = X_5 = k, \qquad X_2 = X_3 = k^2,$$

with arrow maps

$$X_{1\to 2} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad X_{3\to 2} = I_2, \qquad X_{3\to 4} = (1\ 0), \qquad X_{3\to 5} = (0\ 1).$$

If f is an endomorphism of X, then the two maps out of X_3 force f_3 to be diagonal, the identity map $X_3 \to X_2$ gives $f_2 = f_3$, and the map $X_1 \to X_2$ forces the two diagonal entries to be equal. Thus $\mathrm{End}(X) = k$, so X is indecomposable. Geometrically, X corresponds to the unique arc ρ not in T_Q such that

$$(e(\rho, \gamma_1), \dots, e(\rho, \gamma_5)) = (1, 2, 2, 1, 1).$$

$\square$

Variant 2. *(40 points)* Let Q be a quiver:

$$\begin{tikzcd}
 & & & 4 \\
1 & 2 \arrow[l] & 3 \arrow[l] \arrow[ur] & \\
 & & & 5 \arrow[ul]
\end{tikzcd}$$

a). Construct $P(3)$ and $I(2)$ and using projective and injective resolution calculate $\tau(P(3))$ and $\tau^{-1}(I(2))$, where τ denotes the Auslander-Reiten translation.

b). Construct a triangulation of punctured polygon corresponding to Q and find which arcs with respect to this triangulation correspond to $P(3)$, $I(2)$, $\tau(P(3))$, $\tau^{-1}(I(2))$ and to indecomposable representation of dimension vector $(1, 2, 2, 1, 1)$.

Proof. By the path description of projectives and injectives,

$$\dim P(3) = (1, 1, 1, 1, 0), \qquad \dim I(2) = (0, 1, 1, 0, 1).$$

The nonzero maps in $P(3)$ are identities along $3 \to 2$, $2 \to 1$, and $3 \to 4$; the nonzero maps in $I(2)$ are identities along $5 \to 3$ and $3 \to 2$. Since $P(3)$ is projective and $I(2)$ is injective,

$$\tau P(3) = 0, \qquad \tau^{-1} I(2) = 0.$$

Construct the punctured-pentagon triangulation $T_Q = \{\gamma_1, \dots, \gamma_5\}$ by Schiffler's rule, labeling γ_i by the vertex i. With R the clockwise tagged rotation,

$$P(3) \leftrightarrow R^{-1}\gamma_3, \qquad I(2) \leftrightarrow R\gamma_2.$$

Thus the formal rotated arcs for $\tau P(3)$ and $\tau^{-1} I(2)$ are γ_3 and γ_2, while the actual representations are zero.

For the dimension vector $(1, 2, 2, 1, 1)$ take $X_1 = X_4 = X_5 = k$ and $X_2 = X_3 = k^2$, with maps

$$X_{2\to 1} = (1 \ \ -1), \qquad X_{3\to 2} = I_2, \qquad X_{3\to 4} = (1 \ 0), \qquad X_{5\to 3} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

The endomorphism relations force the common matrix on X_2 and X_3 to be diagonal, and then the maps to X_1 and X_4 force the two diagonal entries to agree. Hence $\mathrm{End}(X) = k$, so X is indecomposable. It is the arc ρ with crossing vector

$$(e(\rho, \gamma_1), \dots, e(\rho, \gamma_5)) = (1, 2, 2, 1, 1).$$

$\square$

Variant 3. *(40 points)* Let Q be a quiver:

$$\begin{tikzcd}
 & & & 4 \arrow[dl] \\
1 \arrow[r] & 2 \arrow[r] & 3 & \\
 & & & 5 \arrow[ul]
\end{tikzcd}$$

a). Construct $P(2)$ and $I(3)$ and using projective and injective resolution calculate $\tau(P(2))$ and $\tau^{-1}(I(3))$, where τ denotes the Auslander-Reiten translation.

b). Construct a triangulation of punctured polygon corresponding to Q and find which arcs with respect to this triangulation correspond to $P(2)$, $I(3)$, $\tau(P(2))$, $\tau^{-1}(I(3))$ and to indecomposable representation of dimension vector $(1,2,2,1,1)$.

Proof. The projective and injective representations are

$$\dim P(2) = (0,1,1,0,0), \qquad \dim I(3) = (1,1,1,1,1).$$

In $P(2)$ the only nonzero arrow map is the identity on $2 \to 3$; in $I(3)$ all four arrow maps are identities. Since $P(2)$ is projective and $I(3)$ is injective,

$$\tau P(2) = 0, \qquad \tau^{-1} I(3) = 0.$$

Let $T_Q = \{\gamma_1, \dots, \gamma_5\}$ be the punctured-pentagon triangulation associated to this orientation. Then

$$P(2) \leftrightarrow R^{-1}\gamma_2, \qquad I(3) \leftrightarrow R\gamma_3.$$

The formal arcs obtained by rotating once are γ_2 and γ_3; these are triangulation arcs, while the corresponding translated representations in $\operatorname{rep} Q$ are zero.

The required indecomposable of dimension vector $(1,2,2,1,1)$ is obtained by taking $X_1 = X_4 = X_5 = k$, $X_2 = X_3 = k^2$, and

$$X_{1\to 2} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad X_{2\to 3} = I_2, \qquad X_{4\to 3} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \qquad X_{5\to 3} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

The two maps into X_3 make the common endomorphism of X_2 and X_3 diagonal in the standard basis, and the map from X_1 forces the two diagonal entries to be equal. Therefore $\operatorname{End}(X) = k$, so X is indecomposable. It corresponds to the unique arc ρ with crossing vector $(1,2,2,1,1)$ against the arcs γ_i. $\square$

Variant 4. *(40 points)* Let Q be a quiver:

$$\begin{tikzcd}
 & & & 4 \\
1 & 2 \arrow[l] \arrow[r] & 3 \arrow[ur] & \\
 & & & 5 \arrow[ul]
\end{tikzcd}$$

a). Construct $P(2)$ and $I(3)$ and using projective and injective resolution calculate $\tau(P(2))$ and $\tau^{-1}(I(3))$, where τ denotes the Auslander-Reiten translation.

b). Construct a triangulation of punctured polygon corresponding to Q and find which arcs with respect to this triangulation correspond to $P(2)$, $I(3)$, $\tau(P(2))$, $\tau^{-1}(I(3))$ and to indecomposable representation of dimension vector $(1,2,2,1,1)$.

Proof. The path descriptions give

$$\dim P(2) = (1,1,1,1,0), \qquad \dim I(3) = (0,1,1,0,1).$$

The nonzero maps in $P(2)$ are identities along $2 \to 1$, $2 \to 3$, and $3 \to 4$; the nonzero maps in $I(3)$ are identities along $2 \to 3$ and $5 \to 3$. Hence

$$\tau P(2) = 0, \qquad \tau^{-1} I(3) = 0.$$

If $T_Q = \{\gamma_1, \ldots, \gamma_5\}$ is the corresponding triangulation of the punctured pentagon and R is clockwise tagged rotation, then

$$P(2) \leftrightarrow R^{-1}\gamma_2, \qquad I(3) \leftrightarrow R\gamma_3.$$

Thus the formal arcs for the two translated terms are γ_2 and γ_3, although the actual representations are zero.

For the vector $(1,2,2,1,1)$ take $X_1 = X_4 = X_5 = k$ and $X_2 = X_3 = k^2$ with

$$X_{2\to 1} = (1\ \ -1), \qquad X_{2\to 3} = I_2, \qquad X_{3\to 4} = (1\ 0), \qquad X_{5\to 3} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

The same endomorphism calculation as above gives $\mathrm{End}(X) = k$, so X is indecomposable. In the punctured pentagon it is the unique arc ρ satisfying

$$(e(\rho, \gamma_1), \ldots, e(\rho, \gamma_5)) = (1,2,2,1,1).$$

□

Variant 5. *(40 points)* Let Q be a quiver:

$$1 \longleftarrow 2 \longleftarrow 3 \longrightarrow 4 \longleftarrow 5$$

a). Compute the Auslander-Reiten quiver for Q. Calculate $\mathrm{Hom}(P(3), I(4))$.

b). Construct a triangulation of polygon corresponding to Q and find which arcs with respect to this triangulation correspond to $P(3)$, $I(4)$, $\tau(P(3))$, $\tau^{-1}(I(4))$ and to indecomposable representation of dimension vector $(1,1,1,1,1)$.

Proof. Write X_{ab} for the interval representation supported on $a, a+1, \ldots, b$. The vertices of the Auslander-Reiten quiver are all X_{ab} with $1 \le a \le b \le 5$, and its irreducible arrows are

$$X_{11} \to X_{12}, \quad X_{12} \to X_{14}, X_{22}, \quad X_{13} \to X_{23},$$
$$X_{14} \to X_{15}, X_{24}, \quad X_{15} \to X_{13}, X_{25}, \quad X_{22} \to X_{24},$$
$$X_{23} \to X_{33}, \quad X_{24} \to X_{25}, X_{34}, \quad X_{25} \to X_{23}, X_{35},$$
$$X_{34} \to X_{35}, \quad X_{35} \to X_{33}, X_{55}, \quad X_{44} \to X_{14}, X_{45}, \quad X_{45} \to X_{15}.$$

Here $P(3) = X_{14}$ and $I(4) = X_{35}$. Since $\mathrm{Hom}(P(i), M) \cong M_i$ naturally,

$$\mathrm{Hom}(P(3), I(4)) \cong I(4)_3 \cong k,$$

so $\dim \mathrm{Hom}(P(3), I(4)) = 1$.

Number the vertices of an 8-gon clockwise. A triangulation corresponding to Q is

$$T_1 = (1,3), \quad T_2 = (3,8), \quad T_3 = (3,7), \quad T_4 = (4,7), \quad T_5 = (4,6).$$

A diagonal not in the triangulation represents the interval whose support is exactly the set of triangulation diagonals that it crosses. Thus

$$P(3) = X_{14} \leftrightarrow (2,6), \qquad I(4) = X_{35} \leftrightarrow (5,8), \qquad X_{15} \leftrightarrow (2,5).$$

In $\mathrm{rep}\, Q$, $\tau P(3) = 0$ and $\tau^{-1} I(4) = 0$. Under the formal polygon rotation, these boundary positions are the triangulation diagonals $T_3 = (3,7)$ and $T_4 = (4,7)$, respectively. $\square$

Variant 6. *(40 points)* Let Q be a quiver:

$$1 \longrightarrow 2 \longrightarrow 3 \longleftarrow 4 \longleftarrow 5$$

a). Compute the Auslander-Reiten quiver for Q. Calculate $\mathrm{Hom}(P(1), I(3))$.

b). Construct a triangulation of polygon corresponding to Q and find which arcs with respect to this triangulation correspond to $P(1)$, $I(3)$, $\tau(P(1))$, $\tau^{-1}(I(3))$ and to indecomposable representation of dimension vector $(1,1,1,1,1)$.

Proof. Again let X_{ab} denote the interval representation supported on $a, \dots, b$. The Auslander-Reiten quiver has vertices X_{ab}, $1 \le a \le b \le 5$, and irreducible arrows

$$\begin{gathered} X_{12} \to X_{11}, \quad X_{13} \to X_{14}, \quad X_{14} \to X_{15}, X_{44}, \quad X_{15} \to X_{12}, X_{45}, \\ X_{22} \to X_{12}, \quad X_{23} \to X_{13}, X_{24}, \quad X_{24} \to X_{14}, X_{25}, \\ X_{25} \to X_{15}, X_{22}, \quad X_{33} \to X_{23}, X_{34}, \\ X_{34} \to X_{24}, X_{35}, \quad X_{35} \to X_{25}, \quad X_{44} \to X_{45}, \quad X_{45} \to X_{55}. \end{gathered}$$

We have $P(1) = X_{13}$ and $I(3) = X_{15}$. Therefore

$$\mathrm{Hom}(P(1), I(3)) \cong I(3)_1 \cong k,$$

and the Hom space has dimension 1.

A corresponding triangulation of the clockwise numbered 8-gon is

$$T_1 = (1,3), \quad T_2 = (1,4), \quad T_3 = (1,5), \quad T_4 = (5,8), \quad T_5 = (5,7).$$

The relevant diagonals are

$$P(1) = X_{13} \leftrightarrow (2,8), \qquad I(3) = X_{15} \leftrightarrow (2,6), \qquad X_{15} \leftrightarrow (2,6).$$

The actual translated representations are $\tau P(1) = 0$ and $\tau^{-1} I(3) = 0$; the formal rotated arcs are $T_1 = (1,3)$ and $T_3 = (1,5)$. $\square$

Variant 7. *(40 points)* Let Q be a quiver:

$$1 \longrightarrow 2 \longrightarrow 3 \longrightarrow 4 \longleftarrow 5$$

a). Compute the Auslander-Reiten quiver for Q. Calculate $\mathrm{Hom}(P(1), I(4))$.

b). Construct a triangulation of polygon corresponding to Q and find which arcs with respect to this triangulation correspond to $P(1)$, $I(4)$, $\tau(P(1))$, $\tau^{-1}(I(4))$ and to indecomposable representation of dimension vector $(1, 1, 1, 1, 1)$.

Proof. Let X_{ab} be the interval representation supported on $a, \dots, b$. The Auslander-Reiten quiver has irreducible arrows

$$\begin{gathered} X_{12} \to X_{11}, \quad X_{13} \to X_{12}, \quad X_{14} \to X_{15}, \quad X_{15} \to X_{13}, X_{55}, \\ X_{22} \to X_{12}, \quad X_{23} \to X_{13}, X_{22}, \quad X_{24} \to X_{14}, X_{25}, \\ X_{25} \to X_{15}, X_{23}, \quad X_{33} \to X_{23}, \quad X_{34} \to X_{24}, X_{35}, \\ X_{35} \to X_{25}, X_{33}, \quad X_{44} \to X_{34}, X_{45}, \quad X_{45} \to X_{35}. \end{gathered}$$

Here $P(1) = X_{14}$ and $I(4) = X_{15}$, so

$$\mathrm{Hom}(P(1), I(4)) \cong I(4)_1 \cong k.$$

Thus $\dim \mathrm{Hom}(P(1), I(4)) = 1$.

One triangulation of the clockwise numbered 8-gon corresponding to this orientation is

$$T_1 = (1, 3), \quad T_2 = (1, 4), \quad T_3 = (1, 5), \quad T_4 = (1, 6), \quad T_5 = (6, 8).$$

The relevant diagonals are

$$P(1) = X_{14} \leftrightarrow (2, 8), \qquad I(4) = X_{15} \leftrightarrow (2, 7), \qquad X_{15} \leftrightarrow (2, 7).$$

The actual translated representations are zero: $\tau P(1) = 0$ and $\tau^{-1} I(4) = 0$. The formal polygon rotations land on the triangulation diagonals $T_1 = (1, 3)$ and $T_4 = (1, 6)$. □

Common Problems

Choose up to three problems from this list to submit.

1. *(20 points)* Prove that P is projective if and only if $\mathrm{Ext}^1(P, N) = 0$ for all representations N.

Proof. If P is projective, every short exact sequence

$$0 \to N \to E \to P \to 0$$

splits, since the identity map of P lifts through $E \twoheadrightarrow P$. Hence every extension of P by N is trivial, so $\mathrm{Ext}^1(P, N) = 0$.

Conversely, assume that $\mathrm{Ext}^1(P, N) = 0$ for every N. Let $g : M \twoheadrightarrow X$ be an epimorphism and let $f : P \to X$ be a map. Form the pullback

$$E = M \times_X P = \{(m, p) \mid g(m) = f(p)\}.$$

Then

$$0 \to \ker g \to E \to P \to 0$$

is exact. Its class lies in $\operatorname{Ext}^1(P, \ker g) = 0$, so it splits. A section $P \to E$, followed by the projection $E \to M$, gives a lift of f through g. Thus $\operatorname{Hom}(P, -)$ sends epimorphisms to epimorphisms, and P is projective. □

2. *(20 points)* Let $M \in \operatorname{rep} Q$ and

$$0 \to P_1 \to P_0 \to M \to 0$$

be standard projective resolution. Show that

$$0 \to D(M) \to D(P_0) \to D(P_1) \to 0$$

is the injective resolution for $D(M) \in \operatorname{rep} Q^{\mathrm{op}}$. Show that duality functor D takes a minimal projective resolution for M to a injective envelope for $D(M)$.

Proof. Write $D = \operatorname{Hom}_k(-, k)$. Since all vector spaces are finite-dimensional, D is exact and contravariant. Applying D to

$$0 \to P_1 \to P_0 \to M \to 0$$

gives an exact sequence

$$0 \to D(M) \to D(P_0) \to D(P_1) \to 0.$$

For every vertex i, the dual of the projective $P_Q(i)$ is the injective $I_{Q^{\mathrm{op}}}(i)$: a path $i \to j$ in Q is the same as a path $j \to i$ in Q^{op} after reversing arrows and dualizing. Hence $D(P_0)$ and $D(P_1)$ are injective representations of Q^{op}. Therefore the displayed sequence is an injective resolution of $D(M)$.

If the projective resolution is minimal, then the map $P_0 \twoheadrightarrow M$ is a projective cover. Its dual $D(M) \hookrightarrow D(P_0)$ is an essential monomorphism into an injective object: equivalently, if $a : D(P_0) \to D(P_0)$ satisfies $a|_{D(M)} = \mathrm{id}_{D(M)}$, then dualizing gives an endomorphism of P_0 over M, hence an automorphism by minimality of the projective cover. Thus $D(M) \hookrightarrow D(P_0)$ is an injective envelope, and the dual complex is the minimal injective resolution. □

3. *(20 points)* Let $g : M \to N$ be a surjective morphism between representations of Q, and let $P(i)$ be the projective representation at vertex i. Show that the map

$$g_* : \operatorname{Hom}(P(i), M) \to \operatorname{Hom}(P(i), N)$$

is surjective.

Proof. For every representation X there is a natural isomorphism

$$\operatorname{Hom}(P(i), X) \cong X_i, \qquad f \mapsto f_i(e_i),$$

where e_i is the trivial path at i. Under this identification, the map g_* is exactly the component map $g_i : M_i \to N_i$. Since g is surjective, each component g_i is surjective. Hence g_* is surjective. □

4. *(20 points)* Show that for any Q the representations $I(i)$, $i \in Q_0$ are indecomposable.

Proof. In the finite acyclic setting, the dual Yoneda isomorphism gives

$$\mathrm{Hom}(X, I(i)) \cong D(X_i)$$

naturally in X. Taking $X = I(i)$ yields

$$\mathrm{End}(I(i)) \cong D(I(i)_i).$$

Since the only path from i to i is the trivial path, $I(i)_i \cong k$, and therefore $\mathrm{End}(I(i)) \cong k$. If $I(i) = A \oplus B$ with both A and B nonzero, projection onto one summand would give a nonzero idempotent different from 1 in $\mathrm{End}(I(i))$, impossible in the field k. Thus $I(i)$ is indecomposable. □

5. *(20 points)* Show that the smallest wild tree with valence at most 3 at each vertex has 7 vertices.

Proof. For acyclic quivers, representation type depends only on the underlying graph: Dynkin trees are finite type, Euclidean trees are tame, and all other connected trees are wild.

No tree with at most five vertices is wild: a path is of type A_n, and under the maximum-valence-3 condition the only branched possibilities are D_4 and D_5. With six vertices and maximum valence at most 3, the possible trees are

$$A_6, \qquad D_6, \qquad E_6, \qquad \widetilde{D}_5,$$

which are Dynkin or Euclidean. Hence no such tree with at most six vertices is wild.

Now consider the seven-vertex tree with edges

$$ua, \quad ub, \quad uv, \quad vc, \quad vw, \quad wd.$$

It has maximum valence 3: the vertices u and v have valence 3, the vertex w has valence 2, and a, b, c, d are leaves. It is not Dynkin of type A, D, E, and it is not Euclidean: it is not a cycle, not one of the exceptional $\widetilde{E}$ diagrams, and it is not a $\widetilde{D}_n$ diagram because the two trivalent vertices are adjacent while one outer arm has length 2. Therefore it is wild. Thus the smallest possible number of vertices is 7. □